OPTICS CONTINUUM

Photolithography-free switchable broadband thermal emitter based on In_3SbTe_2 for radiative thermal management

MINGZE LI,[1,2] XIQIAO HUANG,[2] BIYUAN WU,[1,2] YAO HONG,[3] LONG WANG,[4,5] AND XIAOHU WU[1,6]

[1] Thermal Science Research Center, Shandong Institute of Advanced Technology, Jinan 250100, Shandong, China
[2] School of Power and Energy, Northwestern Polytechnical University, Xi'an 710072, Shaanxi, China
[3] Transportation Safety Institute, China Academy of Safety Science and Technology, 100012, Beijing, China
[4] Zhijian Laboratory, Rocket Force University of Engineering, Xi'an 710025, Shaanxi, China
[5] wlxs520@126.com
[6] xiaohu.wu@iat.cn

Abstract: Radiative cooling, as a thermal management technology that requires no additional energy input, has gained widespread attention in recent years. It relies on the material's ability to radiatively dissipate heat within the specific atmospheric window (8-14 μm) band. The emission of traditional materials is typically a constant, which cannot meet the need for dynamic control. This paper investigates a photolithography-free switchable broadband thermal emitter based on In_3SbTe_2 (IST) for radiative thermal management. By designing an IST/Al_2O_3/SiO_2 structure, the phase-change characteristics of IST are utilized to switch the emission state. In the "off" state, the emission is approximately 0.15, while in the "on" state, it achieves broadband high emission of around 0.86 in the atmospheric window, with an emission modulation of about 0.71. Additionally, the electric field distribution at the emission peak in the "on" state is analyzed to explore the interaction between electromagnetic waves and the layers. The influence of layer thickness, incident wave polarization, and incident angle on performance is studied, and room-temperature infrared thermal imaging simulations are conducted to visually present the emission variation under different states. The structure demonstrates significant advantages in broadband high emission and tunability, showing promising potential for improving device reliability, extending lifespan, and reducing energy consumption.

1. Introduction

Any object with a temperature above absolute zero emits thermal radiation. Radiative cooling technology, which requires no additional energy input, relies on the material's high emission characteristics in the specific atmospheric transparent window (8-14 μm) band to transfer most of the heat to the cold outer space (around 3 K) [1–8]. Traditional thermal radiation materials typically have a constant emission, but in applications like radiative cooling, although static micro-nano structures (such as gratings [9], photonic crystals [10,11], and metamaterials [12–15]) can achieve spectrally selective emission, they often fail to meet the demand for dynamic emission control [16,17]. Therefore, switchable thermal emitters with "on" and "off" functionalities have increasingly attracted attention [18]. These functionalities are primarily achieved through mechanisms such as thermochromism [19,20], electrochromism [21,22], and mechanical deformation [23–25]. Among them, thermochromism utilizes phase-change materials (PCMs) to realize adaptive emission modulation. These materials exhibit different optical properties at different temperatures due to phase transitions, offering fast response and long-term stability, making them widely applied in various scenarios.

#554500 https://doi.org/10.1364/OPTCON.554500
Journal © 2025 Received 13 Jan 2025; revised 3 Mar 2025; accepted 4 Mar 2025; published 24 Mar 2025

The most commonly used phase-change material for dynamic emission control in the atmospheric window (8-14 μm) is vanadium dioxide (VO_2), which undergoes a phase transition at 68°C [26–28]. For example, Wu et al. [29] designed an intelligent radiative device (SRD) for spacecraft based on VO_2, with an optimized emission modulation of up to 0.64 while maintaining a solar absorption of 0.29. However, the phase transition of VO_2 is volatile, and to maintain its radiative properties in a certain state, the temperature must be strictly controlled to stay above/below the phase transition temperature. In contrast, non-volatile PCMs such as $Ge_2Sb_2Te_5$ (GST) can undergo reversible phase transitions at ambient temperature, with the phase being stable under normal conditions unless triggered by external factors [30–32]. In_3SbTe_2 (IST) is another non-volatile PCMs that can undergo phase transitions locally triggered by a laser. After the phase transition is completed, IST can maintain the post-transition state even after the external stimulus is removed, unaffected by the room temperature environment [33,34]. It exhibits metallic behavior in its crystalline phase and a constant dielectric constant in the amorphous phase. Due to its higher stability at high temperatures and advantages in long-wave infrared thermal emission control, IST has gradually gained attention [35–38].

In recent years, many researchers have achieved radiative control of infrared radiation by designing multilayer structures based on IST [39–41]. For example, Conrads et al. [42] designed a perfect infrared absorbing metasurface structure based on IST, achieving nearly perfect absorption in the 6 to 14 μm band. However, due to the requirement of micro-nano-scale photolithography in the fabrication process, it cannot be scaled up for large-scale production. Giteau et al. [43] designed and fabricated a switchable narrowband diffusive thermal emitter structure. In the amorphous phase, it achieved narrowband high emission in the 9-10 μm range, while in the crystalline phase, it exhibited low emission in the long-wave infrared (LWIR) region. Additionally, The MgF_2/SiC/Si_3N_4/GST/MgF_2/Ag structure and the HfO_2/GST/SiC/GST/MgF_2/Ag structure proposed by Emrose et al. exhibit good emission modulation in the 1-1.6 μm and 2.1-4.1 μm wavelength ranges, respectively [44]. However, passive radiative cooling requires high emission across the entire atmospheric window, and research on photolithography-free switchable broadband thermal emitters based on IST for radiative cooling is still relatively limited.

In this study, we investigate the emission of single-layer and double-layer ENZ materials (SiO_2, Al_2O_3) in the atmospheric window band and design a photolithography-free switchable broadband thermal emitter based on IST/Al_2O_3/SiO_2. Among them, SiO_2 and Al_2O3 have good thermal stability and are less affected by phase transition factors. In the "off" state, the emission of the emitter is approximately 0.15, while in the "on" state (when IST transitions from crystalline to amorphous), it exhibits broadband high emission (approximately 0.86) in the atmospheric window band, with a modulation of up to 0.71 between the two states. The electric field distributions at the emission peaks in the "on" state are analyzed to further explore the interaction between the incident electromagnetic waves and the materials in each layer. Additionally, the effects of the thickness of each layer, the polarization state of the incident wave, and the incident angle on the performance of the emitter are investigated. Infrared thermal imaging simulations at room temperature provide a more intuitive understanding of the emission changes between the "on" and "off" states. This structure demonstrates exceptional performance in the atmospheric window band, with significant advantages in broadband high emission and tunability, making it highly promising for future applications in optoelectronic devices, infrared thermal imaging, energy-saving technologies, and thermal control.

2. Model and method

As shown in Fig. 1(a), the switchable thermal emitter consists of the phase change material IST, along with a thermal emitter composed of an Al_2O_3/SiO_2 double-layer epsilon-near-zero (ENZ) material. Figure 1(b) and Fig. 1(c) display the transmission of IST in the atmospheric window band in both of its phases. When the thickness of IST ranges from 0.02 to 0.1 μm, IST has a high reflection in its crystalline phase and exhibits broadband low emissivity when it is located on the top layer. In the amorphous phase, IST has relatively high transparency, allowing light to pass through smoothly. Arranging the Al_2O_3 and SiO_2 layers beneath it can achieve broadband high emission within the atmospheric window. The dielectric constants of the materials used are shown in Figs. 1(d - f). The dielectric constant of IST is based on the data provided by Hebler et al [45]. The dielectric constants of both Al_2O_3 and SiO_2 are taken from the experimental results reported by Jan Kischkat et al [46].

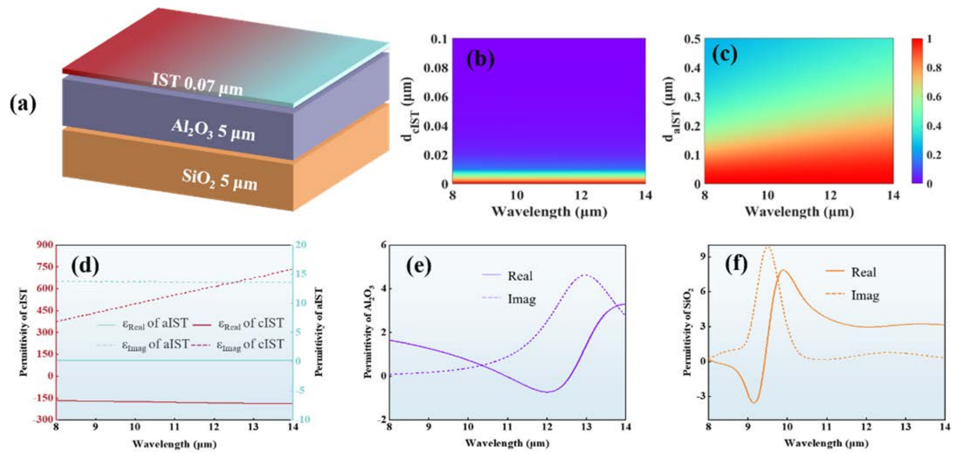

Fig. 1. (a) Schematic diagram of the structure of the switchable thermal emitter. (b) Transmission spectra of IST in the crystalline phase and (c) in the amorphous phase. Dielectric constant of (d) IST in both phases, (e) Al_2O_3 and (f) SiO_2.

In this work, based on Maxwell's equations, we use the transfer matrix method (TMM) to calculate the transmission, reflection, and emission in multilayer planar structures [47,48]. Assuming the plane of incidence is the x-z plane, there is no polarization conversion between the TM and TE modes. The incident wave is assumed to be a TM wave with an incident angle θ, and the electromagnetic field can be expressed as follows:

$$H_y = U_y(z)e^{(-jk_x x)}, \tag{1}$$

$$E_y = j\left(\frac{\mu_0}{\varepsilon_0}\right)^{\frac{1}{2}} S_x(z)e^{(-jk_x x)}, \tag{2}$$

in which $k_x = k_0\sin\theta$, $k_0 = 2\pi/\lambda$, λ is the wavelength. In conjunction with Maxwell's equations, we can obtain the following equations:

$$\frac{\partial}{\partial z'}\begin{bmatrix} U_y \\ S_x \end{bmatrix} = \begin{bmatrix} \frac{j\varepsilon_{xz}}{\varepsilon_{zz}K_x} & \frac{\varepsilon_{xx}-\varepsilon_{zx}\varepsilon_{xz}}{\varepsilon_{zz}} \\ \frac{K_x^2}{\varepsilon_{zz}-1} & \frac{j\varepsilon_{zx}}{\varepsilon_{zz}K_x} \end{bmatrix}\begin{bmatrix} U_y \\ S_x \end{bmatrix}, \tag{3}$$

in which $z' = zk_0$, $K_x = k_x/k_0$. Assuming that the eigenmatrix of the coefficient matrix of Eq. (4) is $W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix}$. Thus, the electromagnetic field can also be written as:

$$\begin{bmatrix} U_y \\ S_x \end{bmatrix} = c^+\begin{bmatrix} w_{11} \\ w_{21} \end{bmatrix} e^{(k_0 q_1 z)} + c^-\begin{bmatrix} w_{12} \\ w_{22} \end{bmatrix} e^{(k_0 q_2 z)}, \tag{4}$$

in which, c^+ and c^- are unknown constants. According to Eqs. (4) and (5), the electromagnetic field in each layer of the planar structure can be described. Then, by applying boundary conditions, the reflection and transmission coefficients can be calculated [49]. Afterward, the spectral directional absorption and emission of the structure can be calculated using Eqs. (5) and (6) [50,51].

$$\alpha(\theta,\lambda) = 1 - R(\theta,\lambda) - T(\theta,\lambda), \tag{5}$$

$$e(\theta,\lambda) = 1 - R(-\theta,\lambda) - T_b(\theta,\lambda), \tag{6}$$

in which, $R(\theta, \lambda)$ and $R(-\theta, \lambda)$ represent the reflections for incident angles θ and $-\theta$ at wavelength λ. $T(\theta, \lambda)$ denotes the transmission for an incident angle θ at wavelength λ, while $T_b(\theta, \lambda)$ refers to the transmission from the backward direction (where the incident wave propagates from right to left) at wavelength λ.

The structure of Al_2O_3/ SiO_2 is used as the thermal emitter because the ENZ wavelengths of SiO_2 are located at 8.1 µm and 9.5 µm respectively, and those of Al_2O_3 are located at 11 µm and 12.6 µm respectively. It can cover most of the long-wave infrared (LWIR) spectrum and achieve high broadband emission within the atmospheric window. The thicknesses of Al_2O_3 and SiO_2 are set to $H_2 = 5$ µm and $H_3 = 5$ µm respectively. The IST should have a high reflection in the crystalline phase and also meet the thickness requirement for the phase transition triggered by laser excitation. Hence, the thickness H_1 of IST is chosen to be 0.07 µm.

3. Results and discussion

In this design, the simulated emission spectra are calculated using both the transfer matrix method (TMM) and COMSOL. Since the thermal emitter is a multilayer planar structure composed entirely of isotropic materials, the results for transverse magnetic (TM) wave incidence and transverse electric (TE) wave incidence are identical under normal incidence. Therefore, we use TM wave incidence as an example for the simulation. As shown in Fig. 2(a), the emission spectra of the single-layer SiO_2, single-layer Al_2O_3, and double-layer Al_2O_3/ SiO_2 structures were calculated respectively. The emission within a specific wavelength range at temperature T can be ascertained by integrating the emission features of the material with blackbody radiation, which is presented by the following formula:

$$\varepsilon = \frac{\int_{\lambda_1}^{\lambda_2} \varepsilon_\lambda E_{b\lambda}(\mathrm{T}) d\lambda}{\int_{\lambda_1}^{\lambda_2} E_{b\lambda}(\mathrm{T}) d\lambda}, \tag{7}$$

$$E_{b\lambda} = \frac{2\pi \mathrm{h_p c^2}}{\lambda^5} \times \frac{1}{e^{h_p/\mathrm{k}\lambda\mathrm{T}} - 1}, \tag{8}$$

in which $E_{b\lambda}$ corresponds to the spectral emission of a blackbody at temperature T. The parameters λ_1 and λ_2 serve to define the lower and upper boundaries of the specific wavelength range. ε_λ represents the emission of the object. As for the symbols, T represents the temperature, $\mathrm{h_p}$ pertains to Planck's constant, c denotes the speed of light in a vacuum, and k is the Boltzmann constant.

The emissions of the single-layer SiO_2 and single-layer Al_2O_3 in the atmospheric window are 0.51 and 0.65 respectively. Moreover, the high emission band of Al_2O_3 is located in the range of 10 - 14 µm, and that of SiO_2 is located in the range of 8 - 10 µm. Combining the two layers can broaden the high emission band. The emission of the Al_2O_3/SiO_2 structure in the atmospheric window can reach 0.82, indicating that this structure has the characteristic of high broadband emission. Figure 2(b) shows that after combining this structure with the IST, the switchable thermal emission (the "on" and "off" states) within the atmospheric window band can be achieved by changing the phase state of IST. The yellow shading represents the atmospheric

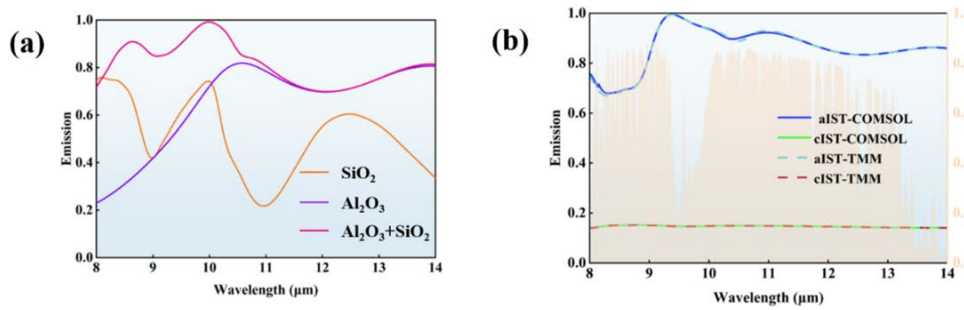

Fig. 2. (a) Emission spectra of SiO_2, Al_2O_3, and Al_2O_3/SiO_2. (b) Emission spectra of the thermal emitter in two states. The yellow shading represents the atmospheric transmission spectrum.

transmission spectrum. We used both TMM method and COMSOL to simulate the emissivity of the switchable broadband thermal emitter, and the results from both methods are in good agreement. When IST is in the crystalline phase (the “off” state), the emission of the emitter is about 0.15. While IST is in the amorphous phase (the “on” state), an emission peak of 0.99 is reached at 9.39 μm, and the overall structure can achieve a high broadband emission (about 0.86). The emission modulation reaches 0.71. The orange shaded area in the figure represents the atmospheric transmission spectrum. To improve its daytime radiative cooling performance, a ZnO-PE solar reflector, fabricated by Cui’s team [52], can be placed on top of the switchable broadband thermal emitter. The solar reflector exhibits an average reflection greater than 0.9 in the sunlight spectrum and an average transmission of 0.8 in the mid-infrared range. It functions by reflecting sunlight while allowing infrared emission to pass through [53]. By integrating the reflector with our designed structure, we aim to achieve the following: (1) The ZnO-PE structure effectively prevents sunlight from reaching the tunable broadband emitter due to its high solar reflection, thereby avoiding heat accumulation. (2) The high tansmission of the ZnO-PE structure in the long-wave infrared range ensures that the radiative heat transfer between the tunable broadband emitter and the atmosphere is not affected.

In order to better understand the interaction between electromagnetic waves and the layers in the switchable thermal emitter, Fig. 3 shows the simulated electric field distribution of the switchable thermal emitter in the “on” state at the resonance wavelengths of 8 μm and 9.38 μm corresponding to the emission peaks. At the wavelength of 8 μm, the electric field is mainly distributed in Al_2O_3 and SiO_2, generating periodic oscillations. While at the wavelength of 9.38 μm, the electric field is mainly concentrated in aIST and Al_2O_3. Therefore, the switchable thermal emitter can effectively absorb the incident light in the “on” state.

The broadband high emission of the switchable thermal emitter depends on the variation in thickness. When IST is in the crystalline phase, it acts as a reflective layer, preventing incident light from passing through to the underlying structure. Therefore, we only investigate the relationship between the thickness of each layer and the emission when IST is in the amorphous phase. Figure 4 shows the emission spectra of the switchable thermal emitter with different layer thicknesses. Obviously, the emission characteristics of this structure, including the position and intensity of the resonant wavelength, are closely related to the thickness of each layer of materials. As shown in Fig. 4(a), when H_1 changes from 0 to 0.5 μm, the emission at wavelengths between 8 - 12 μm gradually decreases, while the emission at wavelengths between 12 - 14 μm gradually increases. Therefore, to ensure broadband high emission within the atmospheric window, H_1 should be controlled to be less than 0.1 μm. As shown in Fig. 4(b), when H_2 is greater than 2 μm, the emission of the structure can maintain a relatively high level, and as the thickness increases, the emission shows periodic enhancement within the wavelength range of 8 -

OPTICS CONTINUUM

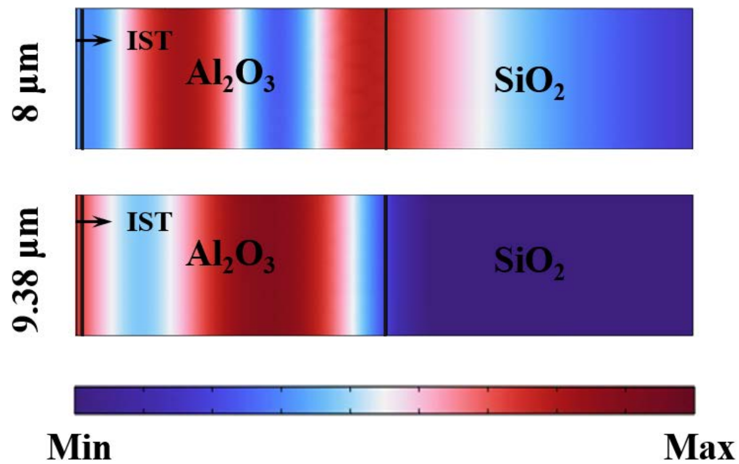

Fig. 3. Electric field distributions of the multilayer thermal emitter structure at 8 μm and 9.38 μm.

10 μm. Figure 4(c) demonstrates that the thickness of H_3 exhibits excellent robustness, ensuring consistent performance within a relatively wide tolerance range. However, it is worth noting that when H_3 is greater than 4 μm, a relatively narrow resonant emission can be observed near the wavelength of 8 μm. Therefore, the finally selected thicknesses for each layer are $H_1 = 0.07$ μm, $H_2 = 5$ μm, and $H_3 = 5$ μm.

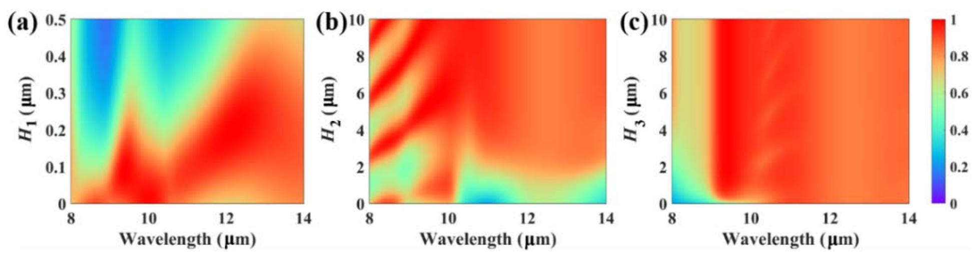

Fig. 4. Emission spectra as functions of the thickness of (a) the aIST, (b) the middle Al_2O_3 and (c) the bottom SiO_2.

Furthermore, the polarization state of the incident light and the incident angle also have a certain impact on the broadband high emission performance. Consequently, an investigation is carried out to explore the correlation between the emission of the switchable thermal emitter within the atmospheric window and the incident angles under both transverse electric (TE) and transverse magnetic (TM) wave conditions, which is depicted in Fig. 5(a) and Fig. 5(b). The results show that under TM wave incidence, the structure is insensitive to the incident angle and always maintains a relatively high value. There is a trough in the emission near 11 μm between 60° and 80°, but the emission will increase near 9 μm and 13 μm. As for the incidence of TE waves, the overall emission decreases slightly, and the high emission is mainly concentrated in the wavelength range between 8 μm and 11 μm. However, it still remains in a situation of broadband high emission overall. Therefore, it can be seen that this structure is less affected by the polarization state and the incident angle.

In addition, to more intuitively demonstrate the changes in emission caused by the "on" and "off" states, infrared thermal imaging simulations were conducted on the structure. As shown in Fig. 5(c), the circular region represents the laser-excited area, while the rest of the structure

OPTICS CONTINUUM

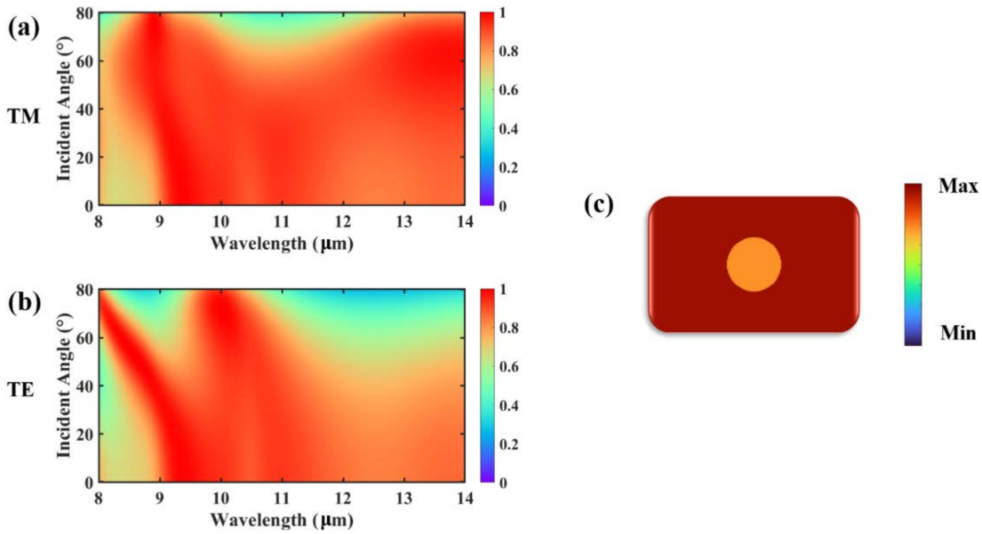

Fig. 5. Regarding the incidence of (a) TM wave and (b) TE wave, the emission of the amorphous phase varies with wavelength and angle (between 0 and 80 degrees). (c) Simulated infrared thermal imaging of this switchable thermal emitter. The circular area is the crystalline phase IST, and the rest of the area is the amorphous phase IST.

remains unexcited. The ambient temperature is set to 27 °C, and the initial temperature of the switchable thermal emitter is set to 40 °C. It is clearly observed that the crystalline phase region exhibits a lower surface temperature (approximately 29 °C) due to its lower emission (about 0.15), showing high reflection similar to metal characteristics. In contrast, the amorphous phase region, with a higher emission (about 0.86), displays a higher surface temperature (approximately 39 °C). This clearly demonstrates the significant effect of different phases on thermal radiation and surface temperature, showcasing the potential of using emission to control surface thermal properties. The calculation formula for the radiation temperature T_{rad} and emission(ε) is:

$$\varepsilon_0 T_{rad}^4 = \varepsilon \tau T_{obj}^4 + (1 - A)\tau T_{sur}^4, \tag{9}$$

here, ε_0 represents the emission of the 8-14 μm infrared thermal imager, typically taken as 1. The value of τ in the 8-14 μm atmospheric window band is assigned as 1 [54–56]. In a state of thermal equilibrium, A is numerically equal to ε.

4. Conclusions

We have designed a switchable broadband thermal emitter operating in the atmospheric window band, based on phase-change material IST. The structure is composed of three layers, IST/Al_2O_3/SiO_2, which avoids the need for complex lithography techniques and allows for large-area fabrication. The phase transition of the top IST layer can be induced by laser. When IST is in the crystalline phase (“off” state), the emission of the emitter is about 0.15. In contrast, when IST is in the amorphous phase (“on” state), the structure exhibits broadband high emission (about 0.86). The emission modulation between the two states reaches 0.71. The electric field distribution at the peak emission in the “on” state is analyzed to explain the physical mechanism behind the broadband high-emission performance. Additionally, the impact of layer thickness on broadband emission is investigated. The effects of polarization state and incident angle on the performance are also examined, showing that the structure is insensitive to both polarization state and incident angle. Finally, infrared thermal imaging simulations are conducted for both states. This switchable broadband thermal emitter demonstrates excellent performance and holds great potential for applications, particularly in ultra-high-temperature thermal management and radiative cooling, providing strong support for the development of more efficient and intelligent technologies.

Funding. National Natural Science Foundation of China (52106099); Natural Science Foundation of Shandong Province (ZR2022YQ57); Taishan Scholars Program.

Acknowledgements. The authors acknowledge the support of the National Natural Science Foundation of China (52106099), the Natural Science Foundation of Shandong Province (ZR2022YQ57), and the Taishan Scholars Program.

Disclosures. The authors declare that there is no conflict of interest.

Data Availability. Data will be made available on reasonable request.

References

1. X. Yin, R. Yang, G. Tan, *et al.*, "Terrestrial radiative cooling: using the cold universe as a renewable and sustainable energy source," Science **370**(6518), 786–791 (2020).
2. B. Wang, L. Li, H. Liu, *et al.*, "Switchable daytime radiative cooling and nighttime radiative warming by VO_2," Sol. Energy Mater Sol. Cells **280**, 113291 (2025).
3. M. J. Yoo, K. R. Pyun, Y. Jung, *et al.*, "Switchable radiative cooling and solar heating for sustainable thermal management," Nanophotonics **13**(5), 543–561 (2024).
4. S. Shi, P. Lv, C. Valenzuela, *et al.*, "Scalable bacterial cellulose-based radiative cooling materials with switchable transparency for thermal management and enhanced solar energy harvesting," Small **19**(39), 2301957 (2023).
5. D. Zhao, A. Aili, Y. Zhai, *et al.*, "Radiative sky cooling: fundamental principles, materials, and applications," Appl. Phys. Rev. **6**(2), 021306 (2019).
6. B. Bhatia, A. Leroy, Y. Shen, *et al.*, "Passive directional sub-ambient daytime radiative cooling," Nat. Commun. **9**(1), 5001 (2018).
7. G. Qi, X. Tan, Y. Jiao, *et al.*, "A scalable and aging-resistant film for radiative cooling," Appl. Therm. Eng. **257**, 124310 (2024).
8. Z. Chen, H. Liu, X. Wu, *et al.*, "Progress in the research of directed thermal radiation," Opto-Electron Eng. **51**(9), 240128 (2024).
9. C. Yang, W. Cai, and Z. M. Zhang, "Tailoring full-Stokes thermal emission from twisted-gratings structures," Nanophotonics **13**(5), 803–811 (2024).
10. K. Yao, H. Ma, M. Huang, *et al.*, "Near-perfect selective photonic crystal emitter with nanoscale layers for daytime radiative cooling," ACS Appl. Nano Mater. **2**(9), 5512–5519 (2019).
11. F. Nan, Y. Zhu, H. Wei, *et al.*, "Flexible composite film with artificial opal photonic crystals for efficient all-day passive radiative cooling," Opt. Express **30**(4), 6003–6015 (2022).
12. J. Liu, Y. Wei, Y. Zhong, *et al.*, "Hierarchical gradient structural porous metamaterial with selective spectral response for daytime passive radiative cooling," Adv. Funct. Mater. **34**(45), 2406393 (2024).
13. H. Yin and C. Fan, "Realization of an efficient radiative cooling emitter with double layer inorganic SiO_2 and TiO_2 metamaterial," Results Phys. **45**, 106216 (2023).
14. X. Guo, B. Wu, H. Li, *et al.*, "Strong spinning thermal radiation enabled by germanium-based chiral dielectric metasurface," J. Opt. **26**(9), 095001 (2024).
15. Y. Zhang, Y. Chen, T. Wang, *et al.*, "Ultrahigh performance passive radiative cooling by hybrid polar dielectric metasurface thermal emitters," Opto-Electron Adv. **7**(4), 230194 (2024).
16. A. Parveen, D. Tyagi, V. Laxmi, *et al.*, "Tunable mid-infrared broadband absorption in the atmospheric window of wavelength in 3–5 µm using VO_2 phase transition in a three-layer metamaterial," Opt. Mater. **153**, 115590 (2024).
17. K. Zhang, Q. Hu, B. Wu, *et al.*, "Lithography-free transparent spacecraft smart radiation device based on phase change material VO_2," Opt. Express **32**(20), 35243 (2024).
18. B. Xie, S. Zhang, W. Zhang, *et al.*, "Multistage smart radiator with gradient emittance based on phase change materials VO_2/GST/IST," Appl. Phys. Lett. **124**(1), 011701 (2024).
19. G. Chen, K. Wang, J. Yang, *et al.*, "Printable thermochromic hydrogel-based smart window for all-weather building temperature regulation in diverse climates," Adv. Mater. **35**(20), 2211716 (2023).
20. X. Mei, T. Wang, M. Chen, *et al.*, "A self-adaptive film for passive radiative cooling and solar heating regulation," J. Mater. Chem. A **10**(20), 11092–11100 (2022).
21. Y. Deng, Y. Yang, Y. Xiao, *et al.*, "Ultrafast switchable passive radiative cooling smart windows with synergistic optical modulation," Adv. Funct. Mater. **33**(35), 2301319 (2023).
22. C. Sui, J. Pu, T. Chen, *et al.*, "Dynamic electrochromism for all-season radiative thermoregulation," Nat. Sustain **6**(4), 428–437 (2023).
23. M. Shi, Z. Song, J. Ni, *et al.*, "Dual-mode porous polymeric films with coral-like hierarchical structure for all-day radiative cooling and heating," ACS Nano **17**(3), 2029–2038 (2023).
24. S. Tao, J. Han, Y. Xu, *et al.*, "Mechanically switchable multifunctional device for regulating passive radiative cooling and solar heating," ACS Appl. Mater. Interfaces **15**(13), 17123–17133 (2023).

25. Q. Zhang, Y. Lv, Y. Wang, *et al.*, "Temperature-dependent dual-mode thermal management device with net zero energy for year-round energy saving," Nat. Commun. **13**(1), 4874 (2022).
26. F. J. Morin, "Oxides which show a metal-to-insulator transition at the neel temperature," Phys. Rev. Lett. **3**(1), 34–36 (1959).
27. K. Tang, K. Dong, J. Li, *et al.*, "Temperature-adaptive radiative coating for all-season household thermal regulation," Science **374**(6574), 1504–1509 (2021).
28. G. Boudan, E. Eustache, L. Divay, *et al.*, "Photonic thermal switch with positive differential emissivity," ACS Photonics **12**(1), 338–347 (2025).
29. B. Wu, X. Wu, H. Liu, *et al.*, "An efficient optimization strategy applied to spacecraft smart radiation devices design," Int. J. Therm. Sci. **195**, 108635 (2024).
30. D. Wang, L. Zhao, S. Yu, *et al.*, "Non-volatile tunable optics by design: From chalcogenide phase-change materials to device structures," Mater. Today **68**, 334–355 (2023).
31. M. Wuttig and N. Yamada, "Phase-change materials for rewriteable data storage," Nat. Mater. **6**(11), 824–832 (2007).
32. M. Wang, B. Wu, F. Wu, *et al.*, "Tunable transmission realized with phase change materials," Plasmonics **16**(1), 71–76 (2021).
33. K. Du, Q. Li, Y. Lyu, *et al.*, "Control over emissivity of zero-static-power thermal emitters based on phase-changing material GST," Light: Sci. Appl. **6**(1), e16194 (2017).
34. E. T. Kim, J. Y. Lee, and Y. T. Kim, "Investigation of electrical characteristics of the $In_3Sb_1Te_2$ ternary alloy for application in phase-change memory," Phys. Status Solidi RRL **3**(4), 103–105 (2009).
35. H. Zhu, B. Xie, W. Zhang, *et al.*, "Self-adaptive multistage infrared radiative thermo-optic modulators based on phase-change materials," Photonics **10**(9), 966 (2023).
36. Y. Maeda, H. Andoh, I. Ikuta, *et al.*, "Reversible phase-change optical data storage in InSbTe alloy films," J. Appl. Phys. **64**(4), 1715–1719 (1988).
37. V. L. Deringer, W. Zhang, P. Rausch, *et al.*, "A chemical link between Ge-Sb-Te and In-Sb-Te phase-change materials," J. Mater. Chem. C **3**(37), 9519–9523 (2015).
38. A. Hebler, S. Wahl, P. T. Kristensen, *et al.*, "Nanostructured In_3SbTe_2 antennas enable switching from sharp dielectric to broad plasmonic resonances," Nanophotonics **11**(17), 3871–3882 (2022).
39. W. Zha, Y. Zhu, B. Ma, *et al.*, "Nonvolatile high-contrast whole long-wave infrared emissivity switching based on In_3SbTe_2," ACS Photonics **10**(7), 2165–2172 (2023).
40. A. Hebler, L. Conrads, K. G. Wirth, *et al.*, "Reconfiguring magnetic infrared resonances with the plasmonic phase-change material In_3SbTe_2," ACS Photonics **9**(5), 1821–1828 (2022).
41. S. Zhou, S. Dong, Y. Guo, *et al.*, "Colored thermal camouflage and anti-counterfeiting with programmable In_3SbTe_2 platform," Nanophotonics **13**(6), 945–954 (2024).
42. L. Conrads, N. Honne, A. Ulm, *et al.*, "Reconfigurable and polarization-dependent grating absorber for large-area emissivity control based on the plasmonic phase-change material In_3SbTe_2," Adv. Opt. Mater. **11**(8), 2202696 (2023).
43. M. Giteau, L. Conrads, A. Mathwieser, *et al.*, "Switchable narrowband diffuse thermal emission with an In_3SbTe_2-based planar structure," Laser Photonics Rev. **19**(5), 2401438 (2024).
44. M. T. Emrose, E. L. Payne, C. You, *et al.*, "Broadband switchable infrared absorbers using phase-change materials," Opt. Mater. Express **14**(3), 577–591 (2024).
45. A. Hebler, S. Wah, T. Leuteritz, *et al.*, "In_3SbTe_2 as a programmable nanophotonics material platform for the infrared," Nat. Commun. **12**(1), 924 (2021).
46. J. Kischkat, S. Peters, B. Gruska, *et al.*, "Mid-infrared optical properties of thin films of aluminum oxide, titanium dioxide, silicon dioxide, aluminum nitride, and silicon nitride," Appl. Opt. **51**(28), 6789–6798 (2012).
47. X. Wu, C. Fu, and Z. M. Zhang, "Influence of hBN orientation on the near-field radiative heat transfer between graphene/hBN heterostructures," J. Photonics Energy **9**(03), 1 (2019).
48. J. Wu, Z. Wang, B. Wu, *et al.*, "The giant enhancement of nonreciprocal radiation in Thue-morse aperiodic structures," Opt. Laser Technol. **152**, 108138 (2022).
49. J. Wu, F. Wu, T. Zhao, *et al.*, "Dual-band nonreciprocal thermal radiation by coupling optical Tamm states in magnetophotonic multilayers," Int. J. Therm. Sci. **175**, 107457 (2022).
50. Z. M. Zhang, X. Wu, and C. Fu, "Validity of Kirchhoff's law for semitransparent films made of anisotropic materials," J. Quant. Spectrosc. Radiat. Transfer **245**, 106904 (2020).
51. Y. Wang, M. Wang, X. Wu, *et al.*, "Review of research on nonreciprocal thermal radiation," Opto-Electron Eng. **51**(9), 240154 (2024).
52. L. Cai, A. Y. Song, W. Li, *et al.*, "Spectrally selective nanocomposite textile for outdoor personal cooling," Adv. Mater. **30**(35), 1802152 (2018).
53. B. Li, J. Hu, C. Chen, *et al.*, "Theoretical study of a highly fault-tolerant and scalable adaptive radiative cooler," Nanophotonics **13**(5), 725–736 (2024).
54. P. Wang, J. Su, P. Ding, *et al.*, "Graphene-metal based tunable radiative metasurface for information encryption and anticounterfeiting," Diamond Relat. Mater. **131**, 109548 (2023).
55. J. Gu, H. Wei, F. Ren, *et al.*, "VO_2-based infrared radiation regulator with excellent dynamic thermal management performance," ACS Appl. Mater. Interfaces **14**(2), 2683–2690 (2022).
56. Q. Kang, K. Guo, X. Zhang, *et al.*, "Dynamically manipulating long-wave infrared polarized thermal radiation by a vanadium dioxide metasurface," Opt. Lett. **49**(9), 2485–2488 (2024).